

Business Areas & Work Scopes

I. Green Remodeling Specialized Consulting Field

- 1 Preliminary/basic survey of aged buildings and on-site condition assessments
- 2 On-site measurement and simulation analysis of energy and indoorcomfort performance
- 3 Initial feasibility review and economic indicator-based analysis including ROI
- 4 Data-driven cost-efficiency optimized design alternatives for improving energy and indoor comfort performance
- 5 On-site measurement and analysis based on safety certification for educational facilities

II. Integrated Consulting Field for Eco-friendly Building Certifications

- 1 Green Building Certification | Energy-efficient Eco-friendly Housing
- 2 Zero Energy Building Certification | Health-friendly Housing Construction Standards
- 3 Intelligent Building Certification | Educational Environment Impact Assessment
- 4 CPTED | Barrier-free Environment Certification
- 5 Total Water Pollution Load Management | Low Impact Development (LID) Preliminary Review

III. Energy and Environmental Performance Analysis & Evaluation Field

- 1 Energy performance simulation analysis and evaluation for certifications
- 2 Energy modeling and object information-linked analysis (Energy Audit)
- 3 Environmental performance analysis and report preparation for optimal design and certification
- 4 On-site measurements for civil complaints such as noise and glare
- 5 Environmental performance analysis and report preparation for environmental impact assessments

IV. Eco-friendly Building Research & Technology Field

- 1 Development of guidelines for design, construction, and maintenance in green remodeling
- 2 Research and analysis of new technology trends in the domestic and international eco-friendly sectors
- 3 KCI and SCI-grade publication activities in all areas of energy and environmental performance
- 4 Development of BIM-based integrated analysis technology for environmental performance
- 5 Development of BIM-based data schema for energy performance

People and spaces
become more **comfortable** and **safer**
in an **eco-friendly** way

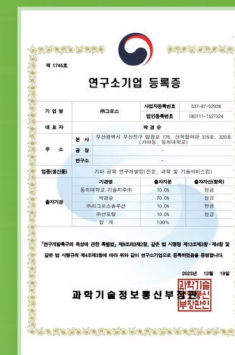
Certificates and Patents



Patent Reg. Cert.

Building energy
simulation device
based on extracted
building shape
information

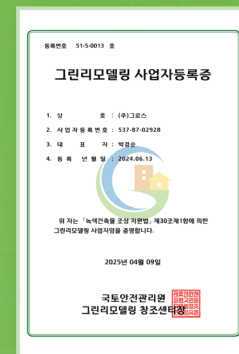
Application Number
10-2020-0142874



Research Institute
Company Reg. Cert.

Research institute
company under the
Special Act on the
Promotion of
Research and
Development Zones

Registration Number
No. 1745



Business Reg. Cert.

Green remodeling
contractor under
the Green Building
Promotion Support
Act

Registration Number
No. 51-5-0013



Measurement
AgencyReg. Cert.

Noise measurement
agency in compliance
with the Environmental
Testing and Inspection
Act

Registration Number
No. JBC-So-38

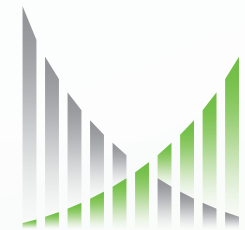


Specialized in eco-friendly low-energy buildings
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Total solution research institute company
based on building environmental performance analysis
R&D | GREEN REMODELING | ENERGY ANALYSIS | ENVIRONMENTAL ANALYSIS



Research Institute Company
GROWTH Co., Ltd.



Eco-friendly green remodeling building doctor

'Green Building Medic Platform'

Solution In progress

A total service covering all stages from on-site precision diagnosis to optimal design, construction, and supervision not just simple remodeling design/construction brokerage



* Each solution(exterior, interior, equipment etc.) can be implemented separately

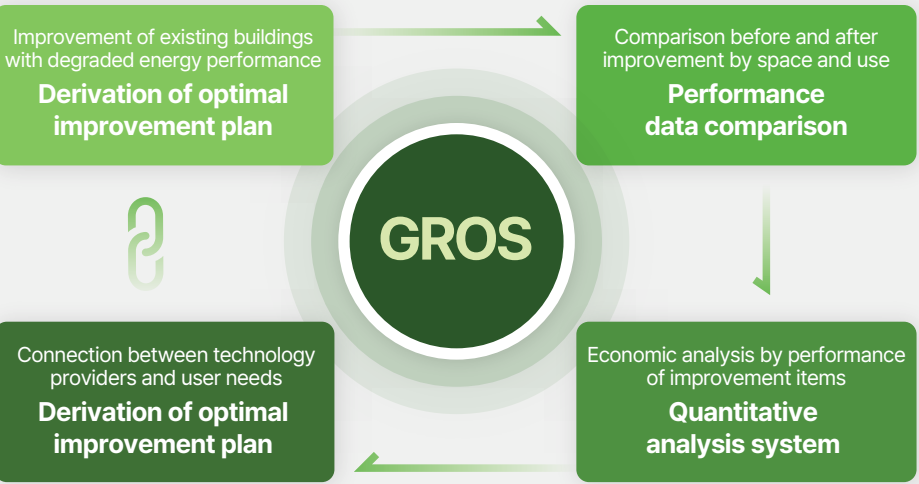
Cost-efficient decision support solution

Development of the green remodeling optimization solution GROS

System Development completed

Development and market deployment of auser-customized green remodeling solution for cost and performance optimization to address blind spots in the private eco-friendly sector

* GROS : Green Remodeling Optimization Solution



Regional energy & carbon reduction solution

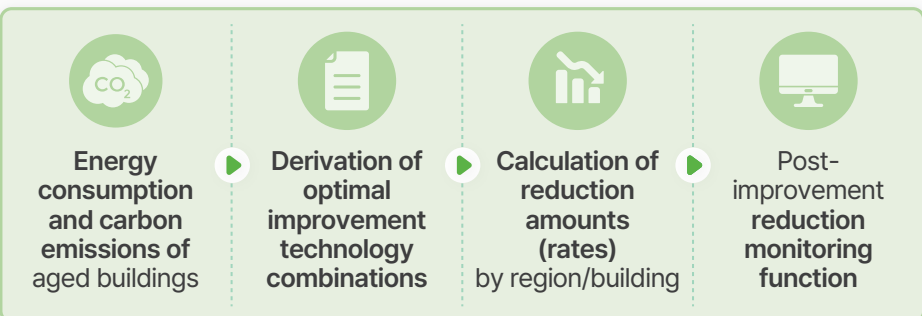
Development of the SGGP solution for analysis, evaluation, and monitoring

R&D In progress

Development of the SGGP solution for reducing building energy consumption and carbon emissions through analysis and monitoring using AI metaheuristic technology

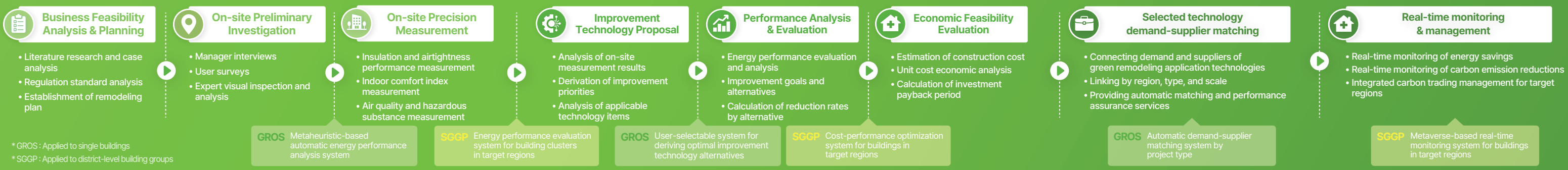
* SGGP : Smart Green City Re-Growth Platform

- ▶ Using AI metaheuristic technology to reduce cost and time without complex data input, and **intuitively monitor the carbon emissions of buildings in the target area**
- ▶ Automatic derivation of energy and carbon reduction factors and associated costs for buildings within the target area, **comprehensively supporting the achievement of policy reduction goals at the administrative district and city levels**



Linked utilization of GROS & SGGP in each stage of green remodeling implementation

Process and key functions · effects



* GROS : Applied to single buildings
* SGGP : Applied to district-level building groups

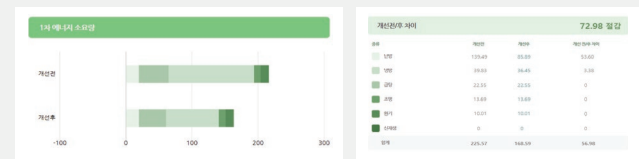
Analysis using 24 types of specialized equipment for precisionon-site measurement (Advanced input data)



Case study verifying the practical applicability of GROS

'Tong-yeong' residential facility project example

GROS energy reduction result : 33.79% reduction (Excluding renewable energy)



Data-based performance diagnosis and reflection of analysis results (Customer-tailored project implemented through direct construction method)



Prototype SGGP solution - UI/UX design (proposal)

